



DC – AI – Routing – Switching – Security – Wireless – Automation

Inline IPSec on MX Series

An HPE Juniper Networking Techpost

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HPE



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1 Inline IPSec on MX Series

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Juniper adds support for inline IPsec on MX-series routers, meaning that IPsec encryption/decryption is done directly by the router's Packet Forwarding Engine (PFE) ASIC instead of by a separate service card, resulting in much higher VPN throughput and lower latency.

This Techpost details how inline IPsec works on Trio 6-based MX routers and describes the configuration steps needed to activate it.

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1.1 Introduction

Internet Protocol security (IPsec) is a protocol suite for securing Internet Protocol (IP) communications by authenticating and encrypting each IP packet of a communication session. IPsec also includes protocols for establishing mutual authentication between agents at the beginning of the session and negotiation of cryptographic keys to be used during the session. Juniper's IPsec implementation supports securing both IPv4 and IPv6 network layers by providing data confidentiality, integrity, origin authentication, replay protection, and non-repudiation of source. The Junos OS combines IPsec with Internet Key Exchange (IKE), which automates the secure generation and management of cryptographic keys and Security Associations (SAs) necessary for establishing encrypted tunnels.

Juniper Inline IPsec is a state-of-the-art feature integrated into Junos OS that offloads IPsec encryption and decryption operations from the CPU to the Packet Forwarding Engine (PFE) ASIC on Juniper routers like the MX304, MX301 or MX10000 Line Cards (based on Trio 6 PFE).

This architectural design enables the MX series routers to deliver hardware-accelerated, high-throughput IPsec VPN capabilities without relying on traditional service cards (such as MS-MPC or SPC3), significantly enhancing performance and reducing latency.

1.2 IPsec Security Association

A Security Association (SA) is a one-way (inbound or outbound) agreement between two communicating peers, that specifies the IPsec protections to be provided to their communications. A typical bidirectional IPsec connection will have two Security Associations, one in each direction. Each SA specifies a Security Parameters Index (SPI), a destination address and the IPsec Encapsulation Protocol. Juniper supports both manual and dynamic SAs. Manual SAs require static configuration of keys and security parameters on both ends. In contrast, dynamic SAs are negotiated using IKE protocols, both IKEv1 and the more modern, flexible IKEv2 are supported. IKE handles mutual authentication, key agreement, and the creation of secure bidirectional communications. Inline IPsec of MX304 supports only IKEv2.

1.3 Encryption and Authentication Algorithms

Juniper IPsec supports strong cryptographic algorithms including AES (Advanced Encryption Standard) for encryption with granular key sizes (128-bit and 256-bit), and SHA-256 for authentication, providing

compliance with high security standards such as FIPS and Common Criteria. The Encapsulating Security Payload (ESP) protocol is used to ensure confidentiality and integrity of the tunneled data while also offering anti-replay protection by default.

1.4 Unified Services Framework

The Unified Services Framework (USF) enables next generation services on MX. This also enables inline IPSec, without relying on external service cards or modules. Enabling USF mode requires a system reboot. Enabling this mode starts various services-related daemons on JUNOS. The CLI configuration for inline services differs between USF mode and non-USF mode, so the router must be in base configuration before enabling this mode and a reboot is required.

The USF can be enabled as follows:

```

1 regress@rtme-mx304-09> show system unified-services status
2 Unified Services : Disabled
3
4 regress@rtme-mx304-09> request system enable unified-services
5 Before enabling unified services, please move to baseline configuration.
6 Are above conditions satisfied ? [yes,no] (no) yes
7
8 Verified junos-unified-services signed by PackageDevelopmentECP256_2025 method ECDSA256+SHA256
9 NOTICE: 'pending' set will be activated at next reboot...
10 Unified-Services upgrade staged. Please reboot with 'request system reboot' command to complete
    the upgrade
11
12 regress@rtme-mx304-09> request system reboot
13 Reboot the system ? [yes,no] (no) yes
14
15 *** FINAL System shutdown message from regress@rtme-mx304-09 ***
16
17 System going down IMMEDIATELY
18
19
20
21 Shutdown NOW!
22 [pid 69698]
```

Note: Make sure that we are triggering the command on both the RE's explicitly

```

1 regress@rtme-mx304-09> show system unified-services status
2 Unified Services : Enabled
```

1.5 SI Interface

A Service Interface (SI) is a virtual physical interface that resides on the Packet Forwarding Engine (PFE) or lookup engine of the ASIC. The SI interface is also called Anchor interface, which enables advanced services like PPPoE, L2TP, MLPPP and IPSec VPN Processing. The SI interface on MX is the anchor for inline cryptographic processing, eliminating the need for separate service cards or PICs.

Configuring the inline-services creates the “si” interfaces. Four “si” interfaces will be created per Trio 6 ASIC, ie. two per PFE.

```

1 regress@rtme-mx304-09> show configuration groups IPSec_T1 chassis
2 fpc 0 {
3     pic 1 {
4         inline-services;
5     }
6 }
7
8 regress@rtme-mx304-09> show interfaces terse |match si-
```

```

9  si-0/1/0 up up
10 si-0/1/1 up up
11 si-0/1/2 up up
12 si-0/1/3 up up

```

We can also specify the bandwidth for the inline-services

```

1  [edit groups IPSec_T1 chassis fpc 0 pic 1]
2  regress@PE1# set inline-services bandwidth ?
3  Possible completions:
4  <bandwidth> Bandwidth reserved for tunnel service
5  100g 100 gigabits per second
6  10g 10 gigabits per second
7  1g 1 gigabit per second
8  200g 200 gigabits per second
9  20g 20 gigabits per second
10 300g 300 gigabits per second
11 30g 30 gigabits per second
12 400g 400 gigabits per second
13 40g 40 gigabits per second
14 50g 50 gigabits per second
15 60g 60 gigabits per second
16 70g 70 gigabits per second
17 800g 800 gigabits per second
18 80g 80 gigabits per second
19 90g 90 gigabits per second

```

Here “si-0/1/0” and “si-0/1/1” for the first PFE and “si-0/1/2” and “si-0/1/3” for the second PFE.

Aspect	Inside Interface	Outside Interface
Traffic State	Cleartext (decrypted)	Encrypted (ESP)
Direction	To/From internal network	To/From external network
Service Domain	Inside	Outside
Typical Unit	.0	.1
IP Protocol	Original IP(TCP/UDP/etc)	ESP (protocol 50)
Routing	Post Decryption Routing	Pre Encryption Routing

1.5.1 ST0 Interface

Secure Tunnel Interface (ST0) is an internal interface used for route-based IPSec VPNs. It is a virtual interface that routes the cleartext traffic towards an IPSec VPN Tunnel. Each logical unit of st0, like st0.1, st0.2, st0.3 etc corresponds to a separate VPN instance.

```

1  regress@PE1> show interfaces st0 terse
2  Interface Admin Link Proto Local Remote
3  st0 up up
4  st0.1 up up inet

```

1.6 Inline IPSec on MX Trio 6

Trio 6 chip has an IPSec engine in the ASIC, which is part of the packet processing chip and does IPSec inline.

Each Trio 6 has two slices, so two engines. Each slice is a PFE, as it is a fabric destination. Each PFE has one IPSec engine. One IPSec engine can handle 300Gbps half-duplex IPSec traffic. Each IPSec Engine provides 1,000 tunnels, so Trio 6 support a total of 2,000 IPSec Tunnels.

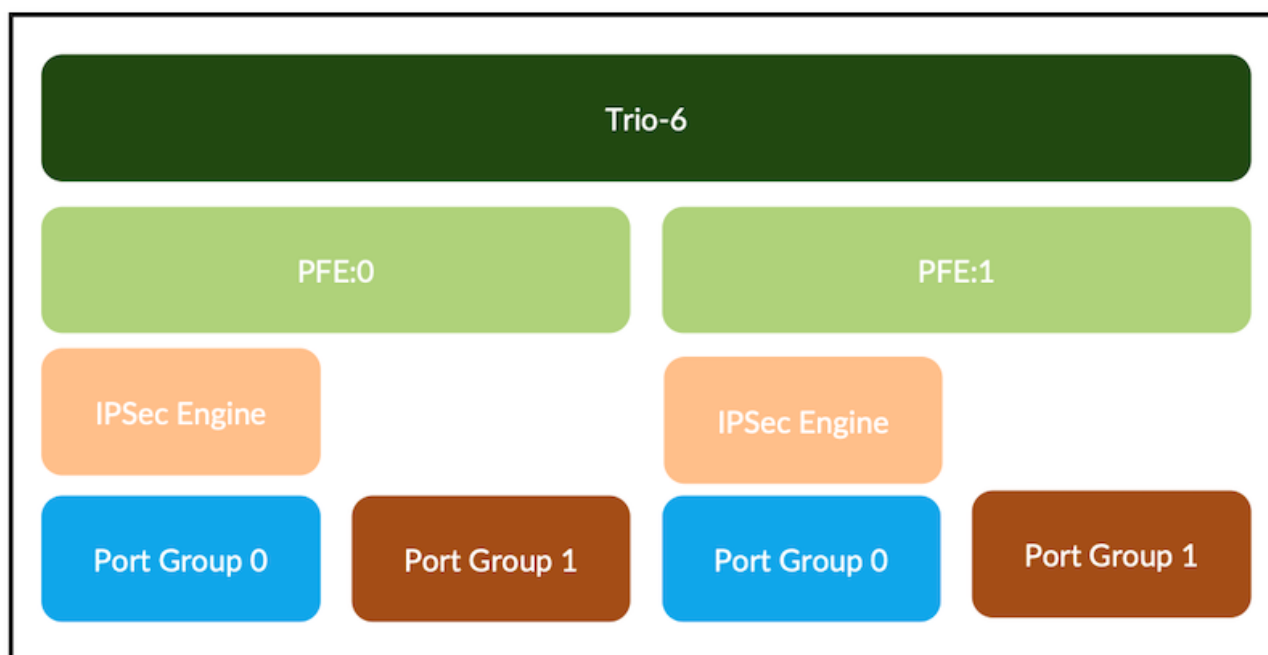


Figure 1: High level view of the Trio 6 PFE

Inline IPSec on Trio 6 based MX platforms (MX304, MX301, LC9600, LC4800, LC4802) works with USF in two modes:

- Traffic Selector mode coming in Junos 25.4 release
- P2P next-hop mode supported since 24.2R1

In P2P mode, routing protocol sees unique st0 logical interface for each tunnel (we will present them as “ifl” in the rest of this document). Protocols add routes with next-hop as st0 ifl. There is one to one mapping with st0 ifl and the tunnel Id. The packet is steered towards the st0 ifl based on route look up in the forwarding path. In this case, each si ifl can be mapped to multiple st0 ifl. In P2P case, the number of tunnels supported is limited by the number of st0 ifl that can be configured in the chassis.

Traffic selection mode configuration is described in the scale tests section of this document.

1.6.1 Encryption Pipeline

In USF P2P mode, there is a static route that points to a st0.x ifl. There is one st0 ifl per tunnel. The packet is steered towards the st0 ifl based on route look up in the forwarding path for encryption. After the route lookup, WAN packets are forwarded to the st0.x interface. They are then passed to the “*si-inside IFL”, which feeds them into the crypto block. Once encrypted, the packets exit through the “si-outside IFL” and continue toward the forwarding next hop.

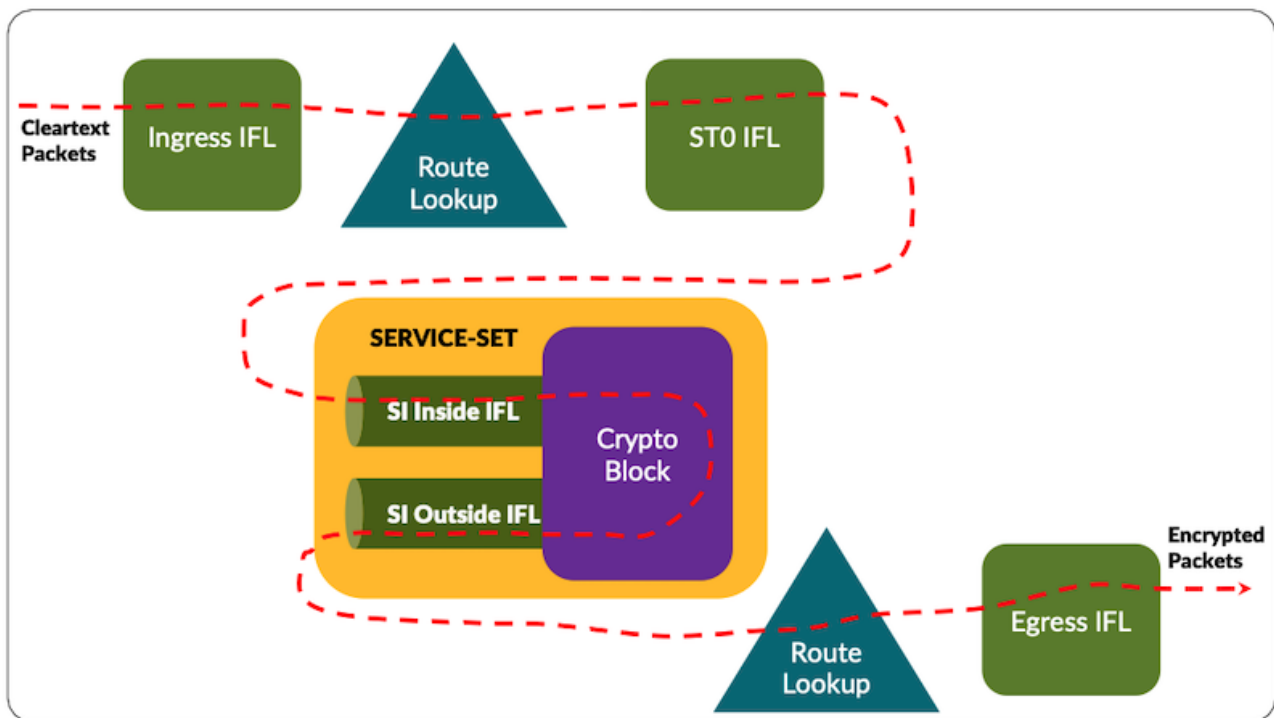


Figure 2: Encryption Pipeline

1.6.2 Decryption Pipeline

In the decryption path, IKE gateway prefix-based route steers the packet towards the outside si ifl.

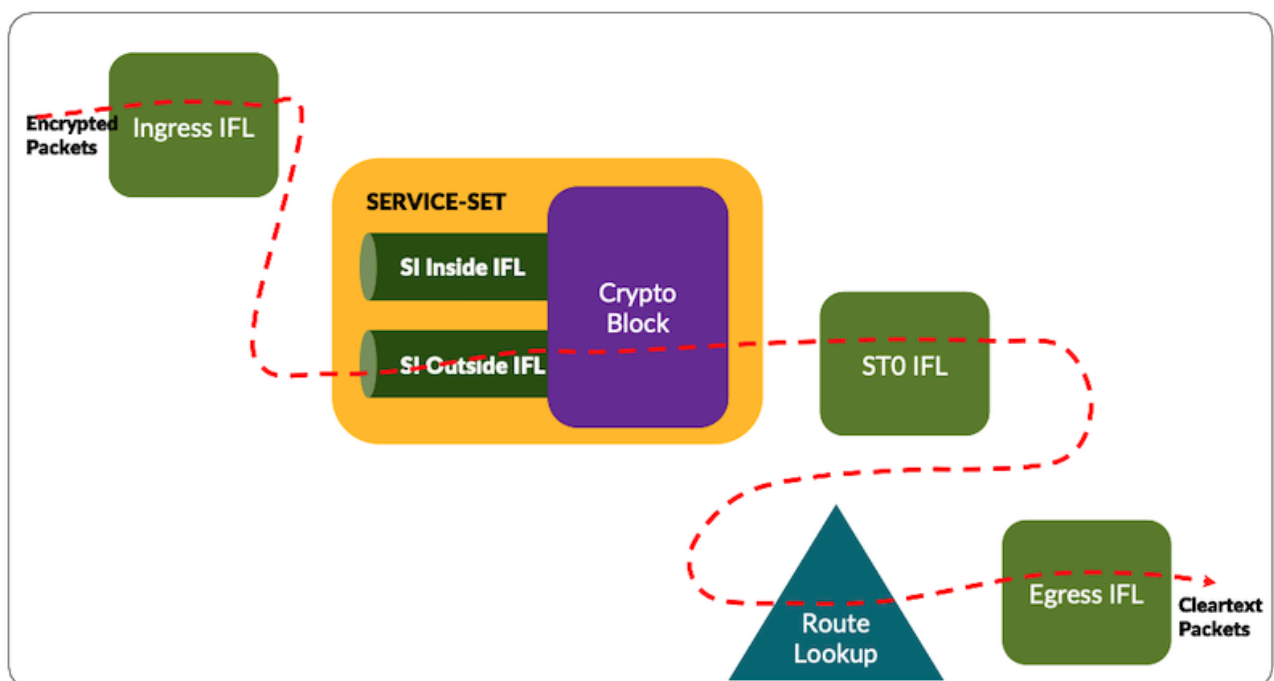


Figure 3: Decryption Pipeline

1.7 Supported Packet Encapsulations via Inline IPSec

The following encapsulation are supported today on Trio 6 MX Series routers or line cards:

- IPv4 over IPSec
- IPv6 over IPSec
- MPLS over GRE over IPSec
- VXLAN over IPSec (vtep and fti)

1.8 Bringing up inline IPSec on MX – P2P

1.8.1 Inline IPSec Configuration Flow

The configuration schema for bringing up inline IPSec on MX304 is as follows:

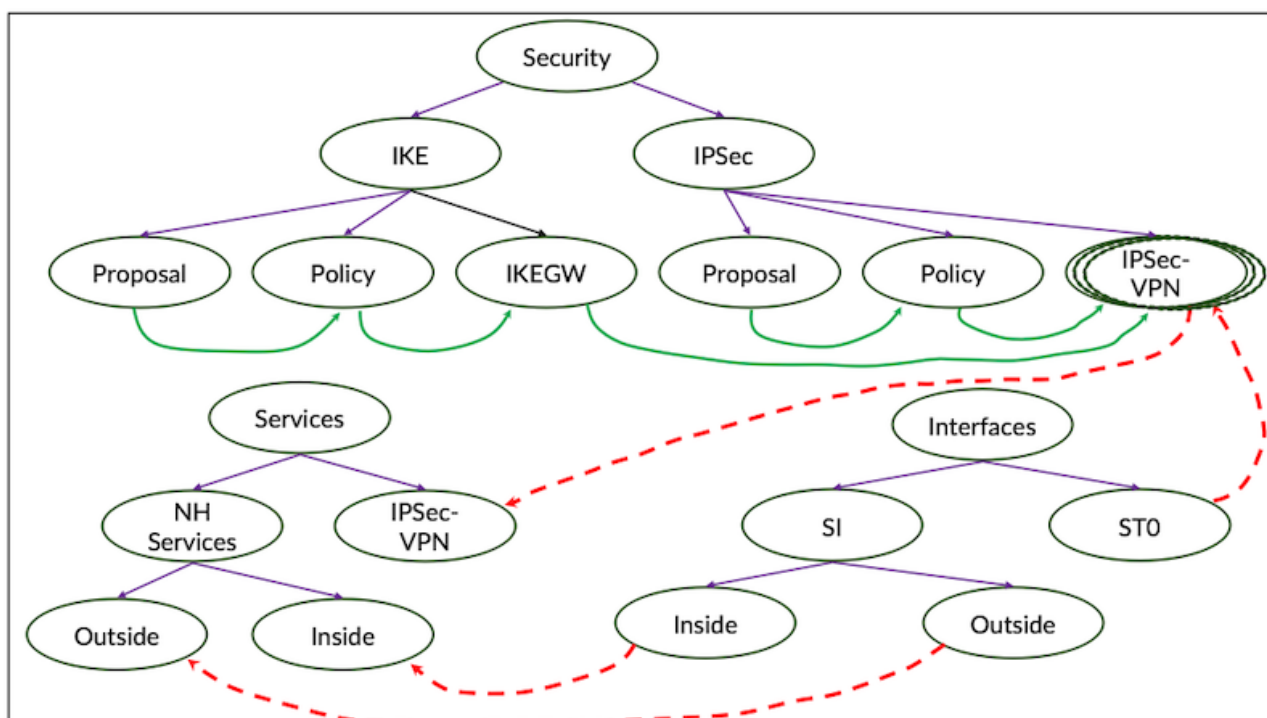


Figure 4: Bringing Up Inline IPSec on MX Router

1.8.2 Test Topology

In the test topology PE routers are having BGP peering for IPv4 and IPv6 family with service provider core running with ISIS SR-MPLS. The CE's are simulated on the traffic generator, PE's and CE's are having EBGP peering for both IPv4 and IPv6 family and CE's advertise 100 routes for each IP family. Traffic is set for both families in bi-directional mode.

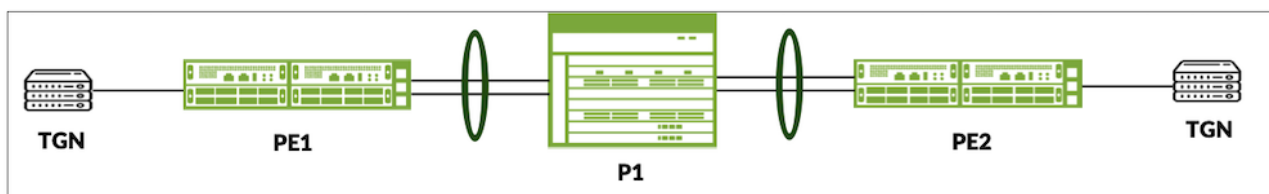


Figure 5: Lab Topology

1.8.3 PE1 Configuration

```

1 regress@PE1> show configuration groups IPSec_Global_v4_v6 |no-more
2 chassis {
3   aggregated-devices {
4     ethernet {
5       device-count 30;
6     }
7   }
8   fpc 0 {
9     pic 1 {
10      inline-services;
11    }
12  }
13  network-services enhanced-ip;
14 }
15 services {
16   service-set SS-1 {
17     next-hop-service {
18       inside-service-interface si-0/1/0.0;
19       outside-service-interface si-0/1/0.1;
20     }
21     ipsec-vpn ipsec_vpn;
22   }
23 }
24 security {
25   ike {
26     proposal ike_proposal {
27       description "IKE Proposal at PE1";
28       authentication-method pre-shared-keys;
29     }
30     policy ike_policy {
31       mode main;
32       proposals ike_proposal;
33       pre-shared-key ascii-text "$9$.fQ3ApBSrv69rvWLVb.P5Ft0SyeKWVw"; ## SECRET-DATA
34     }
35     gateway ike_gw {
36       ike-policy ike_policy;
37       address 13.10.2.2;
38       external-interface ae0;
39       local-address 13.10.1.1;
40       version v2-only;
41     }
42   }
43   ipsec {
44     proposal ipsec_proposal {
45       extended-sequence-number;
46       description "IPSec Proposal at PE1";
47       protocol esp;
48       encryption-algorithm aes-256-gcm;
49     }
50     policy ipsec_policy {
51       proposals ipsec_proposal;
52     }
53     vpn ipsec_vpn {
54       bind-interface st0.1;
55       copy-outer-dscp;
56       ike {
57         gateway ike_gw;
58         anti-replay-window-size 4096;
59         ipsec-policy ipsec_policy;
60       }
61       establish-tunnels immediately;
62     }
63   }

```

```

64 }
65 interfaces {
66   lo0 {
67     unit 0 {
68       family inet {
69         address 12.1.1.1/32;
70       }
71       family iso {
72         address 49.0002.0120.0100.1001.00;
73       }
74       family inet6 {
75         address 2002:12:1:1::1/128;
76       }
77     }
78   }
79   si-0/1/0 {
80     unit 0 {
81       family inet;
82       family inet6;
83       service-domain inside;
84     }
85     unit 1 {
86       family inet;
87       family inet6;
88       service-domain outside;
89     }
90   }
91   st0 {
92     unit 1 {
93       family inet;
94       family inet6;
95     }
96   }
97   et-0/2/10 {
98     description "towards CE1 at IXIA";
99     unit 0 {
100       family inet {
101         address 13.11.1.1/24;
102       }
103       family inet6 {
104         address 2002:13:11:1::1/64;
105       }
106     }
107   }
108   ae0 {
109     description "Bundle to P1";
110     aggregated-ether-options {
111       lacp {
112         active;
113         periodic fast;
114       }
115     }
116     unit 0 {
117       family inet {
118         address 13.10.1.1/24;
119       }
120       family iso;
121       family mpls;
122     }
123   }
124   et-0/2/3 {
125     gigether-options {
126       802.3ad ae0;
127     }
128   }
129   et-0/2/4 {
130     gigether-options {
131       802.3ad ae0;
132     }
133   }
134 }
135 policy-options {
136   policy-statement nhs {
137     term nhs {
138       then {
139         next-hop self;
140       }
141     }
142   }
143   policy-statement pplb {
144     term pplb {

```

```

145         then {
146             load-balance per-flow;
147         }
148     }
149 }
150 }
151 protocols {
152     bgp {
153         group IBGP {
154             type internal;
155             local-address 12.1.1.1;
156             family inet {
157                 unicast;
158             }
159             family inet6 {
160                 labeled-unicast {
161                     inactive: explicit-null;
162                 }
163             }
164             export nhs;
165             neighbor 12.1.1.2;
166         }
167         group EBGP {
168             type external;
169             family inet {
170                 unicast;
171             }
172             peer-as 20001;
173             neighbor 13.11.1.2;
174         }
175         group EBGPv6 {
176             type external;
177             family inet6 {
178                 unicast;
179             }
180             peer-as 20001;
181             neighbor 2002:13:11:1::2;
182         }
183     }
184     isis {
185         interface lo0.0 {
186             passive;
187         }
188         interface ae0.0 {
189             level 1 disable;
190             point-to-point;
191         }
192         source-packet-routing {
193             srgb start-label 20000 index-range 2000;
194             node-segment {
195                 ipv4-index 101;
196                 ipv6-index 201;
197             }
198         }
199         level 1 disable;
200         level 2 wide-metrics-only;
201     }
202     mpls {
203         ipv6-tunneling;
204         interface ae0.0;
205     }
206     lldp {
207         interface et-0/1/4;
208         interface ae0;
209     }
210 }
211 routing-options {
212     rib inet6.0 {
213         static {
214             route ::ffff:12.1.1.2/128 next-hop st0.1;
215         }
216     }
217     router-id 12.1.1.1;
218     autonomous-system 64512;
219     static {
220         route 12.1.1.2/32 next-hop st0.1;
221     }
222     forwarding-table {
223         export pplb;
224     }
225 }

```

1.8.4 PE2 Configuration

```

1 regress@PE2> show configuration groups IPSec_Global_v4_v6 |no-more
2 chassis {
3     aggregated-devices {
4         ethernet {
5             device-count 30;
6         }
7     }
8     fpc 0 {
9         pic 1 {
10             inline-services;
11         }
12     }
13     network-services enhanced-ip;
14 }
15 services {
16     service-set SS-1 {
17         next-hop-service {
18             inside-service-interface si-0/1/0.0;
19             outside-service-interface si-0/1/0.1;
20         }
21         ipsec-vpn ipsec_vpn;
22     }
23 }
24 security {
25     ike {
26         proposal ike_proposal {
27             description "IKE Proposal at PE2";
28             authentication-method pre-shared-keys;
29         }
30         policy ike_policy {
31             mode main;
32             proposals ike_proposal;
33             pre-shared-key ascii-text "$9$fQ3/u0RlK8CtK8X7sYfTz601leMWXNs2"; ## SECRET-DATA
34         }
35         gateway ike_gw {
36             ike-policy ike_policy;
37             address 13.10.1.1;
38             external-interface ae1;
39             local-address 13.10.2.2;
40             version v2-only;
41         }
42     }
43     ipsec {
44         proposal ipsec_proposal {
45             extended-sequence-number;
46             description "IPSec Proposal at PE2";
47             protocol esp;
48             encryption-algorithm aes-256-gcm;
49         }
50         policy ipsec_policy {
51             proposals ipsec_proposal;
52         }
53         vpn ipsec_vpn {
54             bind-interface st0.1;
55             copy-outer-dscp;
56             ike {
57                 gateway ike_gw;
58                 anti-replay-window-size 4096;
59                 ipsec-policy ipsec_policy;
60             }
61             establish-tunnels immediately;
62         }
63     }
64 }
65 interfaces {
66     lo0 {
67         unit 0 {
68             family inet {
69                 address 12.1.1.2/32;
70             }
71             family iso {
72                 address 49.0002.0120.0100.1002.00;
73             }
74         }
75     }
76 }

```

```

74     family inet6 {
75         address 2002:12:1:1::2/128;
76     }
77 }
78 }
79 si-0/1/0 {
80     unit 0 {
81         family inet;
82         family inet6;
83         service-domain inside;
84     }
85     unit 1 {
86         family inet;
87         family inet6;
88         service-domain outside;
89     }
90 }
91 st0 {
92     unit 1 {
93         family inet;
94         family inet6;
95     }
96 }
97 et-0/0/4 {
98     description "towards CE2 at IXIA";
99     unit 0 {
100         family inet {
101             address 13.12.1.1/24;
102         }
103         family inet6 {
104             address 2002:13:12:1::1/64;
105         }
106     }
107 }
108 ae1 {
109     description "bundle to P1";
110     aggregated-ether-options {
111         lacp {
112             active;
113             periodic fast;
114         }
115     }
116     unit 0 {
117         family inet {
118             address 13.10.2.2/24;
119         }
120         family iso;
121         family mpls;
122     }
123 }
124 et-0/0/10 {
125     gigether-options {
126         802.3ad ae1;
127     }
128 }
129 et-0/0/11 {
130     gigether-options {
131         802.3ad ae1;
132     }
133 }
134 }
135 policy-options {
136     policy-statement pplb {
137         term pplb {
138             then {
139                 load-balance per-flow;
140             }
141         }
142     }
143     policy-statement nhs {
144         term nhs {
145             then {
146                 next-hop self;
147             }
148         }
149     }
150 }
151 protocols {
152     bgp {
153         group IBGP {
154             type internal;

```

```

155     local-address 12.1.1.2;
156     family inet {
157         unicast;
158     }
159     family inet6 {
160         labeled-unicast {
161             inactive: explicit-null;
162         }
163     }
164     export nhs;
165     neighbor 12.1.1.1;
166 }
167 group EBGp {
168     type external;
169     family inet {
170         unicast;
171     }
172     peer-as 20002;
173     neighbor 13.12.1.2;
174 }
175 group EBGpV6 {
176     type external;
177     family inet6 {
178         unicast;
179     }
180     peer-as 20002;
181     neighbor 2002:13:12:1::2;
182 }
183 }
184 isis {
185     interface lo0.0 {
186         passive;
187     }
188     interface ae1.0 {
189         level 1 disable;
190         point-to-point;
191     }
192     source-packet-routing {
193         srgb start-label 20000 index-range 2000;
194         node-segment {
195             ipv4-index 102;
196             ipv6-index 202;
197         }
198     }
199     level 1 disable;
200     level 2 wide-metrics-only;
201 }
202 mpls {
203     ipv6-tunneling;
204     interface ae1.0;
205 }
206 lldp {
207     interface et-0/1/4;
208     interface ae1;
209 }
210 }
211 routing-options {
212     rib inet6.0 {
213         static {
214             route ::ffff:12.1.1.1/128 next-hop st0.1;
215         }
216     }
217     router-id 12.1.1.2;
218     autonomous-system 64512;
219     static {
220         route 12.1.1.1/32 next-hop st0.1;
221     }
222     forwarding-table {
223         export pplb;
224     }
225 }

```

1.8.5 P1 Configuration

```

1 regress@P1> show configuration groups IPSec_Global_v4_v6 |no-more
2 chassis {
3     aggregated-devices {

```

```
4     ethernet {
5         device-count 30;
6     }
7 }
8 network-services enhanced-ip;
9 }
10 interfaces {
11     ae0 {
12         description "Bundle to PE1";
13         aggregated-ether-options {
14             lacp {
15                 active;
16                 periodic fast;
17             }
18         }
19         unit 0 {
20             family inet {
21                 address 13.10.1.2/24;
22             }
23             family iso;
24             family mpls;
25         }
26     }
27     et-0/2/0 {
28         gigether-options {
29             802.3ad ae0;
30         }
31     }
32     et-0/2/1 {
33         gigether-options {
34             802.3ad ae0;
35         }
36     }
37     lo0 {
38         unit 0 {
39             family inet {
40                 address 12.1.1.3/32;
41             }
42             family iso {
43                 address 49.0002.0120.0100.1003.00;
44             }
45         }
46     }
47     ae1 {
48         description "Bundle to PE2";
49         aggregated-ether-options {
50             lacp {
51                 active;
52                 periodic fast;
53             }
54         }
55         unit 0 {
56             family inet {
57                 address 13.10.2.1/24;
58             }
59             family iso;
60             family mpls;
61         }
62     }
63     et-0/0/0 {
64         gigether-options {
65             802.3ad ae1;
66         }
67     }
68     et-0/2/2 {
69         gigether-options {
70             802.3ad ae1;
71         }
72     }
73 }
74 policy-options {
75     policy-statement pplb {
76         term pplb {
77             then {
78                 load-balance per-flow;
79             }
80         }
81     }
82 }
83 routing-options {
84     router-id 12.1.1.3;
```

```

85     autonomous-system 64512;
86     forwarding-table {
87         export pplb;
88     }
89 }
90 protocols {
91     isis {
92         interface lo0.0 {
93             passive;
94         }
95         interface ae0.0 {
96             level 1 disable;
97             point-to-point;
98         }
99         interface ae1.0 {
100             level 1 disable;
101             point-to-point;
102         }
103         source-packet-routing {
104             srgb start-label 20000 index-range 2000;
105             node-segment {
106                 ipv4-index 103;
107                 ipv6-index 203;
108             }
109         }
110         level 1 disable;
111         level 2 wide-metrics-only;
112     }
113     mpls {
114         ipv6-tunneling;
115         interface ae0.0;
116         interface ae1.0;
117     }
118     lldp {
119         interface ae0;
120         interface ae1;
121     }
122 }

```

1.8.6 Verification of Inline IPSec

The IKE and IPSec Security Association are as follows:

```

1 regress@PE1> show security ike security-associations
2 Index State Initiator cookie Responder cookie Mode Remote Address
3 75 UP e0b71d9e1c29c324 1ace68c4dcfa427b IKEv2 13.10.2.2
4
5 regress@PE1> show security ipsec security-associations
6 Total active tunnels: 1 Total IPsec sas: 1
7 ID Algorithm SPI Life:sec/kb Mon lsys Port Gateway
8 <500011 ESP:aes-gcm-256/aes256-gcm 0xd0c6331 1222/ unlim - root 500 13.10.2.2
9 >500011 ESP:aes-gcm-256/aes256-gcm 0xe1ed7345 1222/ unlim - root 500 13.10.2.2
10
11 regress@PE1> show security ipsec security-associations detail
12
13 ID: 500011 Virtual-system: root, VPN Name: ipsec_vpn
14 Local Gateway: 13.10.1.1, Remote Gateway: 13.10.2.2
15 Local Identity: ipv4(0.0.0.0-255.255.255.255)
16 Remote Identity: ipv4(0.0.0.0-255.255.255.255)
17 TS Type: proxy-id
18 Version: IKEv2
19 Quantum Secured: No
20 Hardware Offloaded: No
21 PFS group: N/A, Packet Encapsulation: None, Dest port: 0
22 Passive mode tunneling: Disabled
23 DF-bit: clear, Copy-Outer-DSCP: Enabled, Bind-interface: st0.1 , Policy-name: ipsec_policy
24 Port: 500, Nego#: 0, Fail#: 0, Def-Del#: 0 Flag: 0
25 Tunnel events:
26 Mon Nov 17 2025 21:54:47: IPsec SA is deleted because IPsec SA hard life-time expired, sent
27 DEL notification (3 times) <- [repeated sequence END]
28 Mon Nov 17 2025 21:54:47: IPsec SA rekey succeeds (3 times)
29 Mon Nov 17 2025 21:54:47: IPSEC SA rekey initiated because soft life timer expired (3 times)
30 <- [repeated sequence START]
31 Mon Nov 17 2025 19:28:59: IPsec SA is deleted because received DEL notification from peer (1
32 times)
33 Mon Nov 17 2025 19:28:56: IPsec SA rekey succeeds (1 times)

```

```

31 Mon Nov 17 2025 18:40:37: IPsec SA is deleted because IPsec SA hard life-time expired, sent
   DEL notification (4 times) <- [repeated sequence END]
32 Mon Nov 17 2025 18:40:37: IPsec SA rekey succeeds (4 times)
33 Mon Nov 17 2025 18:40:37: IPSEC SA rekey initiated because soft life timer expired (4 times)
   <- [repeated sequence START]
34 Mon Nov 17 2025 15:26:26: IPsec SA is deleted because received DEL notification from peer (2
   times) <- [repeated sequence END]
35 Mon Nov 17 2025 15:26:26: IPsec SA rekey succeeds (2 times) <- [repeated sequence START]
36 Location: FPC 0, PIC 1
37 Anchorship: Thread 0
38 Distribution-Profile: si-0/1/0
39 Direction: inbound, SPI: 0x3160cdce, AUX-SPI: 0
40 , VPN Monitoring: -
41 Hard lifetime: Expires in 715 seconds
42 Lifesize Remaining: Unlimited
43 Soft lifetime: Expires in 12 seconds
44 Mode: Tunnel(0 0), Type: dynamic, State: installed
45 Protocol: ESP, Authentication: aes256-gcm, Encryption: aes-gcm (256 bits)
46 Anti-replay service: counter-based enabled, Replay window size: 4096
47 Extended-Sequence-Number: Enabled
48 tunnel-establishment: establish-tunnels-immediately
49 IKE SA Index: 75
50 Direction: outbound, SPI: 0x36c700ce, AUX-SPI: 0
51 , VPN Monitoring: -
52 Hard lifetime: Expires in 715 seconds
53 Lifesize Remaining: Unlimited
54 Soft lifetime: Expires in 12 seconds
55 Mode: Tunnel(0 0), Type: dynamic, State: installed
56 Protocol: ESP, Authentication: aes256-gcm, Encryption: aes-gcm (256 bits)
57 Anti-replay service: counter-based enabled, Replay window size: 4096
58 Extended-Sequence-Number: Enabled
59 tunnel-establishment: establish-tunnels-immediately
60 IKE SA Index: 75

```

1.8.7 Packet Flow - Encryption

Traffic hitting the Ingress IFL of PE1 destined to hosts behind PE2

```

1 regress@PE1> show interfaces extensive et-0/2/10.0 |match pps
2   Input packets: 160002 10000 pps
3   Output packets: 0 0 pps
4   Input packets: 160002 10000 pps
5   Output packets: 0 0 pps

```

PE2 is reachable via the st0.1 ifl

```

1 regress@PE1> show route 12.1.1.2
2
3 inet.0: 227 destinations, 228 routes (227 active, 0 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
5
6 12.1.1.2/32 *[Static/5] 02:29:23
7   > via st0.1
8   [IS-IS/18] 02:29:26, metric 20
9   > to 13.10.1.2 via ae0.0

```

The network 60.1.1.0/24 behind PE2 is having PNH as that of PE2 which got resolved over st0.1

```

1 regress@PE1> show route 60.1.1.0
2
3 inet.0: 227 destinations, 228 routes (227 active, 0 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
5
6 60.1.1.0/24 *[BGP/170] 02:28:19, localpref 100, from 12.1.1.2
7   AS path: 20002 I, validation-state: unverified
8   > via st0.1

```

The detailed output of the route is as follows:

```

1 regress@PE1> show route 60.1.2.0 extensive
2
3 inet.0: 227 destinations, 228 routes (227 active, 0 holddown, 0 hidden)
4 60.1.2.0/24 (1 entry, 1 announced)

```

```

5 TSI:
6 KRT in-kernel 60.1.2.0/24 -> {indirect(1048574)}
7 Page 0 idx 0, (group EBGp type External) Type 1 val 0x244f4308 (adv_entry)
8   Advertised metrics:
9     Nexthop: Self
10    AS path: [64512] 20002 I
11    Communities:
12    Advertise: 00000001
13 Path 60.1.2.0
14 from 12.1.1.2
15 Vector len 4. Val: 0
16   *BGP Preference: 170/-101
17     Next hop type: Indirect, Next hop index: 0
18     Address: 0x132147fc
19     Next-hop reference count: 200
20     Kernel Table Id: 0
21     Source: 12.1.1.2
22     Next hop type: Router, Next hop index: 577
23     Next hop: via st0.1, selected
24     Session Id: 140
25     Protocol next hop: 12.1.1.2
26     Indirect next hop: 0xadb8388 1048574 INH Session ID: 323, INH non-key opaque: 0x0,
27       INH key opaque: 0x0
28     State: <Active Int Ext>
29     Local AS: 64512 Peer AS: 64512
30     Age: 2:29:24 Metric2: 0
31     Validation State: unverified
32     Task: BGP_64512.12.1.1.2
33     Announcement bits (3): 0-KRT 5-BGP_RT_Background 6-Resolve tree 4
34     AS path: 20002 I
35     Accepted
36     Localpref: 100
37     Router ID: 100.1.1.1
38     Thread: junos-main
39     Indirect next hops: 1
40       Protocol next hop: 12.1.1.2 ResolvState: Resolved TSP index: 4
41       Indirect next hop: 0xadb8388 1048574 INH Session ID: 323, INH non-key opaque:
42         0x0, INH key opaque: 0x0
43       Indirect path forwarding next hops: 1
44         Next hop type: Router
45         Next hop: via st0.1
46         Session Id: 140
47         12.1.1.2/32 Originating RIB: inet.0
48         Node path count: 1
49         Helper node: 0x16b99100
50         Forwarding nexthops: 1
51           Next hop type: Router
52           Next hop: via st0.1
53           Session Id: 140

```

The Global IPv4 packet encryption is as follows:

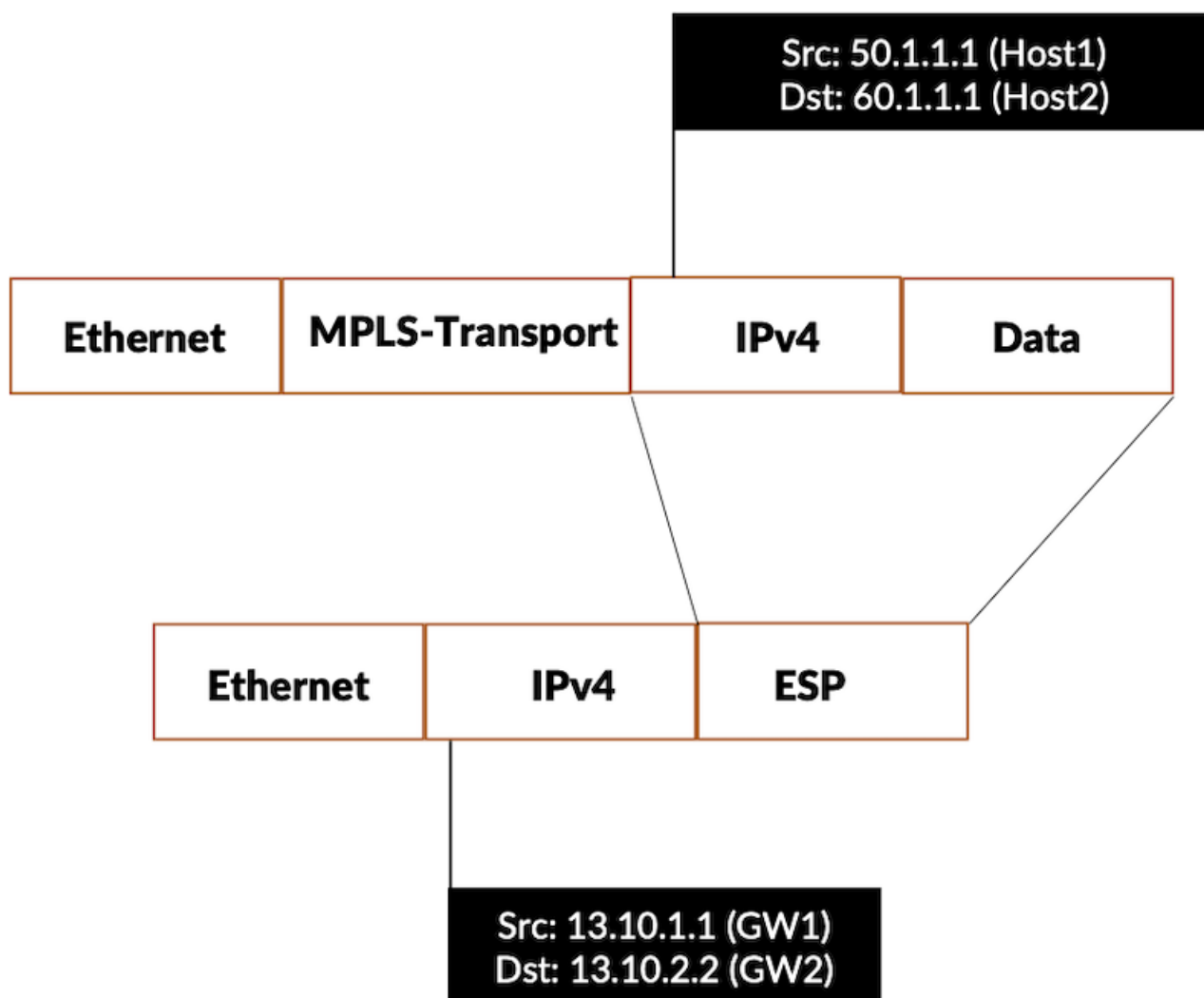


Figure 6: IPv4 packet encryption

The IPv6 Route 3000:60:1:1::1 is as follows:

```
1 regress@PE1> show route 3000:60:1:1::1
2
3 inet6.0: 209 destinations, 209 routes (209 active, 0 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
5
6 3000:60:1:1::/64 *[BGP/170] 3d 16:31:06, localpref 100, from 12.1.1.2
7   AS path: 20002 I, validation-state: unverified
8   > via st0.1, Push 17
```

The detailed output shows the PNH as the IPv4 mapped IPv6 address, which got pointed to the secure tunnel interface

```
1 regress@PE1> show route 3000:60:1:1::1 extensive
2
3 inet6.0: 209 destinations, 209 routes (209 active, 0 holddown, 0 hidden)
4 3000:60:1:1::/64 (1 entry, 1 announced)
5 TSI:
6 KRT in-kernel 3000:60:1:1::/64 -> {indirect(1048575)}
7 Page 0 idx 1, (group EBGpV6 type External) Type 1 val 0x2458ca78 (adv_entry)
8   Advertised metrics:
9     Nexthop: Self
10    AS path: [64512] 20002 I
11   Communities:
12   Advertise: 00000001
13 Path 3000:60:1:1::
14 from 12.1.1.2
15 Vector len 4. Val: 1
```

```

16      *BGP Preference: 170/-101
17      Next hop type: Indirect, Next hop index: 0
18      Address: 0xacf897c
19      Next-hop reference count: 200
20      Kernel Table Id: 0
21      Source: 12.1.1.2
22      Next hop type: Router, Next hop index: 606
23      Next hop: via st0.1, selected
24      Label operation: Push 17
25      Label TTL action: prop-ttl
26      Load balance label: Label 17: None;
27      Label element ptr: 0x13330c60
28      Label parent element ptr: 0x0
29      Label element references: 1
30      Label element child references: 0
31      Label element lsp id: 0
32      Session Id: 16f
33      Protocol next hop: ::ffff:12.1.1.2
34      Label operation: Push 17
35      Label TTL action: prop-ttl
36      Load balance label: Label 17: None;
37      Indirect next hop: 0xadbbf88 1048575 INH Session ID: 378, INH non-key opaque: 0x0,
      INH key opaque: 0x0
38      State: <Active Int Ext>
39      Local AS: 64512 Peer AS: 64512
40      Age: 3d 16:31:10 Metric2: 0
41      Validation State: unverified
42      Task: BGP_64512.12.1.1.2
43      Announcement bits (3): 0-KRT 1-BGP_RT_Background 7-Resolve tree 3
44      AS path: 20002 I
45      Accepted
46      Route Label: 17
47      Localpref: 100
48      Router ID: 100.1.1.1
49      Thread: junos-main
50      Indirect next hops: 1
51      Protocol next hop: ::ffff:12.1.1.2 ResolvState: Resolved TSP index: 3
52      Label operation: Push 17
53      Label TTL action: prop-ttl
54      Load balance label: Label 17: None;
55      Indirect next hop: 0xadbbf88 1048575 INH Session ID: 378, INH non-key opaque:
      0x0, INH key opaque: 0x0
56      Indirect path forwarding next hops: 1
57      Next hop type: Router
58      Next hop: via st0.1
59      Session Id: 16f
60      ::ffff:12.1.1.2/128 Originating RIB: inet6.0
61      Node path count: 1
62      Helper node: 0x16f23280
63      Forwarding nexthops: 1
64      Next hop type: Router
65      Next hop: via st0.1
66      Session Id: 16f

```

The IPv6 Encrypted packet is as follows:

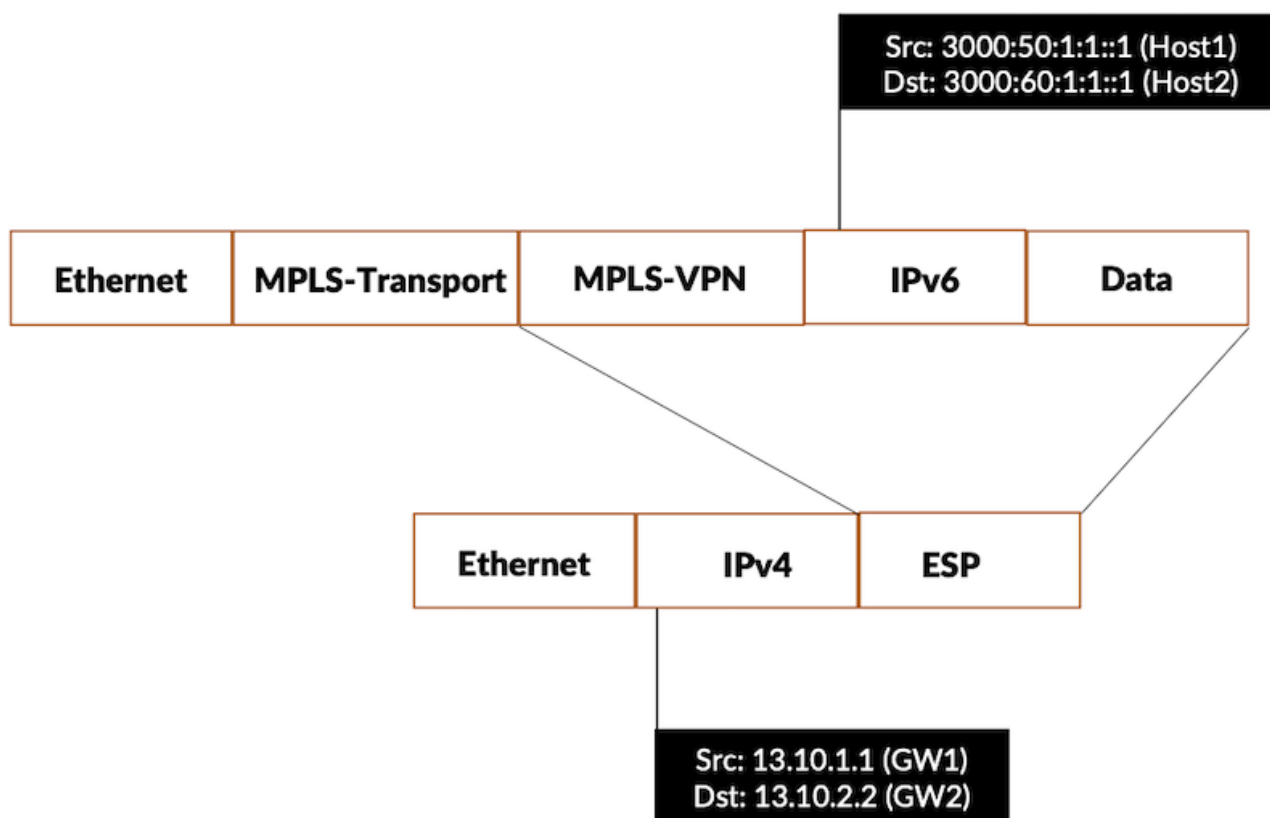


Figure 7: IPv6 packet encryption

The packets hitting the st0.1 is as follows:

```

1 regress@PE1> show interfaces extensive st0.1 |match pps
2   Input packets: 1252 0 pps
3   Output packets: 169943767 10001 pps
4   Input packets: 1252 0 pps
5   Output packets: 169943767 10001 pps
6
7 regress@PE1>

```

The packets are send to Crypto Engine via the “SI-0/1/0.0” inside IFL

```

1 regress@PE1> show interfaces extensive si-0/1/0.0 |match pps
2   Input packets: 4900014 10004 pps
3   Output packets: 0 0 pps
4   Input packets: 4900014 10004 pps
5   Output packets: 0 0 pps
6   Input packets: 0 0 pps
7   Output packets: 0 0 pps

```

From the crypto, the packets are send to the “SI-0/1/0.1” outside IFL

```

1 regress@PE1> show interfaces extensive si-0/1/0.1 |match pps
2   Input packets: 5200081 10001 pps
3   Output packets: 0 0 pps
4   Input packets: 5200081 10001 pps
5   Output packets: 0 0 pps
6   Input packets: 0 0 pps
7   Output packets: 0 0 pps

```

The encrypted packet is sent to the forwarding nexthop

```

1 regress@PE1> show interfaces extensive ae0|match pps
2   Input packets: 1530 2 pps
3   Output packets: 5421597 10006 pps
4   Input packets: 0 0 pps
5   Statistics Packets pps Bytes bps

```

The IPSec stats for the encrypted packet is as follows:

```

1 regress@PE1> show security ipsec statistics
2 ESP Statistics:
3   Encrypted bytes: 19718998
4   Decrypted bytes: 0
5   Encrypted packets: 39917
6   Decrypted packets: 0
7 AH Statistics:
8   Input bytes: 0
9   Output bytes: 0
10  Input packets: 0
11  Output packets: 0
12 Errors:
13   AH authentication failures: 0, Replay errors: 0
14   ESP authentication failures: 0, ESP decryption failures: 0
15   Bad headers: 0, Bad trailers: 0
16
17 regress@PE1>

```

1.8.8 Packet Flow: Decryption

Encrypted packets hitting the WAN port ae1.0

```

1 regress@PE2> show interfaces extensive ae1 |match pps
2   Input packets: 264104739 10005 pps
3   Output packets: 72224 3 pps
4   Input packets: 0 0 pps
5   Statistics Packets pps Bytes bps

```

The long route points to the si outside ifl

```

1 regress@PE2> show interfaces extensive si-0/1/0.1 |match pps
2   Input packets: 264418443 10004 pps
3   Output packets: 0 0 pps
4   Input packets: 264418443 10004 pps
5   Output packets: 0 0 pps
6   Input packets: 0 0 pps
7   Output packets: 0 0 pps

```

On the decryption path, the packet wont take the si inside ifl, hence the packet pps is zero

```

1 regress@PE2> show interfaces extensive si-0/1/0.0 |match pps
2   Input packets: 1954 0 pps
3   Output packets: 0 0 pps
4   Input packets: 1954 0 pps
5   Output packets: 0 0 pps
6   Input packets: 0 0 pps
7   Output packets: 0 0 pps

```

After the decryption, the packets hits the st0.1 interface

```

1 regress@PE2> show interfaces extensive st0.1 |match pps
2   Input packets: 264495210 10005 pps
3   Output packets: 1955 0 pps
4   Input packets: 264495210 10005 pps
5   Output packets: 1955 0 pps
6   Input packets: 0 0 pps
7   Output packets: 0 0 pps

```

The lookup will lead to the forwarding nexthop towards the end destination network

```

1 regress@PE2> show interfaces extensive et-0/0/4.0 |match pps
2   Input packets: 0 0 pps
3   Output packets: 264695191 10005 pps
4   Input packets: 0 0 pps
5   Output packets: 264693294 10005 pps

```

The decrypted packet status is as follows:

```

1 regress@PE2> show security ipsec statistics

```

```

2  ESP Statistics:
3  Encrypted bytes: 0
4  Decrypted bytes: 1573985804
5  Encrypted packets: 0
6  Decrypted packets: 2889110
7  AH Statistics:
8  Input bytes: 0
9  Output bytes: 0
10 Input packets: 0
11 Output packets: 0
12 Errors:
13 AH authentication failures: 0, Replay errors: 0
14 ESP authentication failures: 0, ESP decryption failures: 0
15 Bad headers: 0, Bad trailers: 0
16
17 regress@PE2>

```

1.9 EVPN-VXLAN over IPSec with P2P

EVPN-VXLAN is a modern data center and campus fabric technology that combines Ethernet VPN as the control plane with VXLAN as the overlay data plane to provide scalable Layer 2 and Layer 3 connectivity over an IP underlay network. In this model, VXLAN encapsulates Ethernet frames into UDP packets using a 24-bit VXLAN Network Identifier (VNI), allowing millions of isolated virtual networks that can span across racks, pods, or even multiple sites without relying on traditional spanning tree. EVPN uses MP-BGP to advertise MAC and IP reachability between VXLAN Tunnel Endpoints (VTEPs), enabling control-plane based MAC learning, reduced flooding, and features such as all-active multihoming and distributed anycast gateways for optimized east-west traffic. Together, EVPN and VXLAN deliver a highly scalable, multi-tenant, and resilient overlay fabric that is now a de facto standard for cloud data centers and large enterprise networks.

The VXLAN traffic can be secured with IPSec. In the below topology, EVPN-VXLAN is brought between the PE's and VXLAN traffic is tunneled via the IPSec.

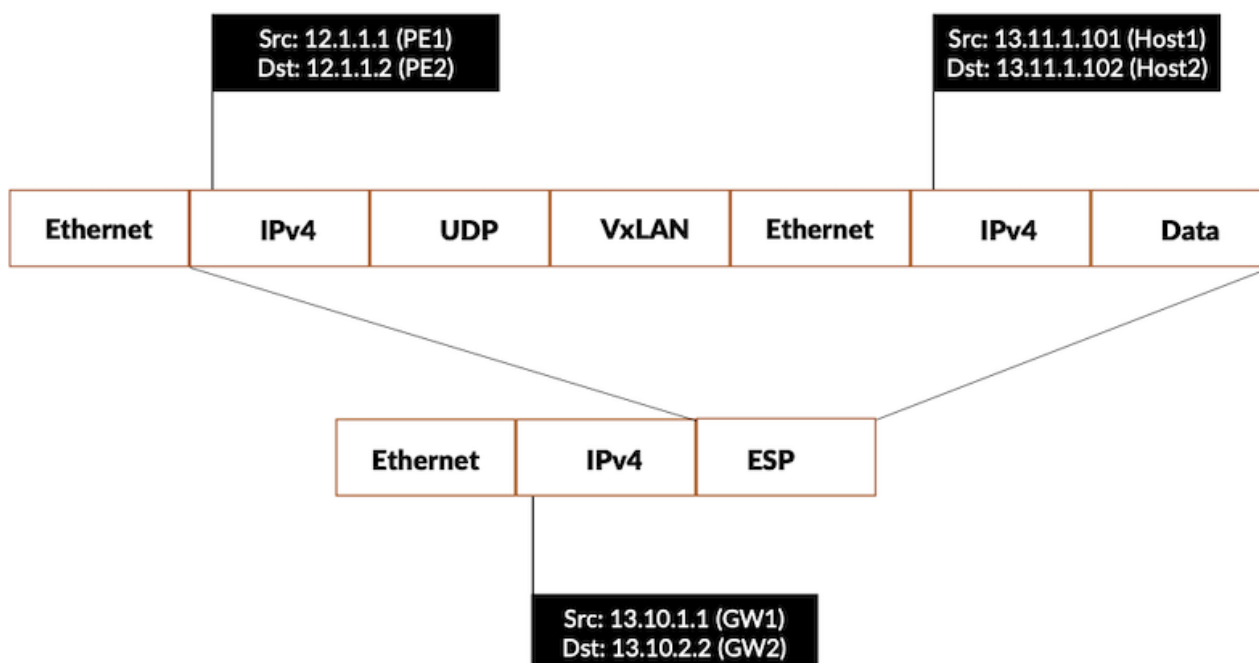


Figure 8: VxLAN traffic tunneled via IPSec

```

> Frame 6934: 259 bytes on wire (2072 bits), 259 bytes captured (2072 bits) on interface unknown, id 0
> Ethernet II, Src: d4:99:6c:92:48:ce (d4:99:6c:92:48:ce), Dst: d4:99:6c:6f:08:50 (d4:99:6c:6f:08:50)
> Internet Protocol Version 4, Src: 12.1.1.1, Dst: 12.1.1.2
> User Datagram Protocol, Src Port: 56379, Dst Port: 4789
> Virtual Extensible Local Area Network, VNI: 10010
> Ethernet II, Src: CETTechn_00:00:01 (00:11:01:00:00:01), Dst: Cisco_00:00:01 (00:12:01:00:00:01)
> Internet Protocol Version 4, Src: 13.11.1.101, Dst: 13.11.1.102
> Data (175 bytes)

> Frame 2000: 259 bytes on wire (2072 bits), 259 bytes captured (2072 bits) on interface unknown, id 0
> Ethernet II, Src: d4:99:6c:92:48:ce (d4:99:6c:92:48:ce), Dst: d4:99:6c:6f:08:50 (d4:99:6c:6f:08:50)
> Internet Protocol Version 4, Src: 13.10.1.1, Dst: 13.10.2.2
  Encapsulating Security Payload
    ESP SPI: 0x0f6b6572 (258696562)
    ESP Sequence: 2144971082

```

Figure 9: Packet Capture

1.9.1 PE1 Configuration

```

1 regress@PE1> show configuration groups IPSec_EVPN_VxLAN |no-more
2 chassis {
3     aggregated-devices {
4         ethernet {
5             device-count 30;
6         }
7     }
8     fpc 0 {
9         pic 1 {
10             inline-services;
11         }
12     }
13     network-services enhanced-ip;
14 }
15 services {
16     service-set SS-EVPN-VxLAN {
17         next-hop-service {
18             inside-service-interface si-0/1/0.0;
19             outside-service-interface si-0/1/0.1;
20         }
21         ipsec-vpn ipsec_evpn_vxlan;
22     }
23 }
24 security {
25     ike {
26         proposal ike_proposal {
27             description "IKE Proposal at PE1";
28             authentication-method pre-shared-keys;
29         }
30         policy ike_policy {
31             mode main;
32             proposals ike_proposal;
33             pre-shared-key ascii-text "$9$.fQ3ApBSrv69rvWLVb.P5Ft0SyeKWVw"; ## SECRET-DATA
34         }
35         gateway ike_gw {
36             ike-policy ike_policy;
37             address 13.10.2.2;
38             external-interface ae0;
39             local-address 13.10.1.1;
40             version v2-only;
41         }
42     }
43     ipsec {
44         proposal ipsec_proposal {
45             extended-sequence-number;
46             description "IPSec Proposal at PE1";
47             protocol esp;
48             encryption-algorithm aes-256-gcm;
49         }

```

```

50     policy ipsec_policy {
51         proposals ipsec_proposal;
52     }
53     vpn ipsec_evpn_vxlan {
54         bind-interface st0.1;
55         copy-outer-dscp;
56         ike {
57             gateway ike_gw;
58             anti-replay-window-size 4096;
59             ipsec-policy ipsec_policy;
60         }
61         establish-tunnels immediately;
62     }
63 }
64 }
65 interfaces {
66     lo0 {
67         unit 0 {
68             family inet {
69                 address 12.1.1.1/32;
70             }
71             family iso {
72                 address 49.0002.0120.0100.1001.00;
73             }
74             family inet6 {
75                 address 2002:12:1:1::1/128;
76             }
77         }
78     }
79     si-0/1/0 {
80         mtu 9192;
81         unit 0 {
82             family inet;
83             family inet6;
84             service-domain inside;
85         }
86         unit 1 {
87             family inet;
88             family inet6;
89             service-domain outside;
90         }
91     }
92     st0 {
93         unit 1 {
94             family inet {
95                 mtu 9192;
96                 address 15.1.1.1/32;
97             }
98             family inet6 {
99                 mtu 9192;
100            }
101        }
102    }
103    et-0/2/10 {
104        description "towards CE1 at IXIA";
105        flexible-vlan-tagging;
106        mtu 9192;
107        encapsulation flexible-ethernet-services;
108        unit 10 {
109            encapsulation vlan-bridge;
110            vlan-id 10;
111        }
112        unit 20 {
113            encapsulation vlan-bridge;
114            vlan-id 20;
115        }
116    }
117    ae0 {
118        description "Bundle to P1";
119        mtu 9192;
120        aggregated-ether-options {
121            lacp {
122                active;
123                periodic fast;
124            }
125        }
126        unit 0 {
127            family inet {
128                address 13.10.1.1/24;
129            }
130            family iso;

```

```

131     family mpls;
132 }
133 }
134 et-0/2/3 {
135     gige-ether-options {
136         802.3ad ae0;
137     }
138 }
139 et-0/2/4 {
140     gige-ether-options {
141         802.3ad ae0;
142     }
143 }
144 irb {
145     mtu 9192;
146     unit 10 {
147         virtual-gateway-accept-data;
148         family inet {
149             address 13.11.1.1/24 {
150                 virtual-gateway-address 13.11.1.254;
151             }
152         }
153     }
154     unit 20 {
155         virtual-gateway-accept-data;
156         family inet {
157             address 13.11.2.1/24 {
158                 virtual-gateway-address 13.11.2.254;
159             }
160         }
161     }
162 }
163 }
164 policy-options {
165     policy-statement nhs {
166         term nhs {
167             then {
168                 next-hop self;
169             }
170         }
171     }
172     policy-statement pplb {
173         term pplb {
174             then {
175                 load-balance per-flow;
176             }
177         }
178     }
179 }
180 routing-instances {
181     evpn_vxlan_1 {
182         instance-type mac-vrf;
183         protocols {
184             evpn {
185                 encapsulation vxlan;
186                 default-gateway no-gateway-community;
187             }
188         }
189         vtep-source-interface lo0.0;
190         bridge-domains {
191             V_10 {
192                 vlan-id 10;
193                 interface et-0/2/10.10;
194                 routing-interface irb.10;
195                 vxlan {
196                     vni 10010;
197                 }
198             }
199             V_20 {
200                 vlan-id 20;
201                 interface et-0/2/10.20;
202                 routing-interface irb.20;
203                 vxlan {
204                     vni 10020;
205                 }
206             }
207         }
208         service-type vlan-aware;
209         route-distinguisher 12.1.1.1:1;
210         vrf-target target:64512:1;
211     }

```

```

212     vrf_1 {
213         instance-type vrf;
214         interface irb.10;
215         interface irb.20;
216         route-distinguisher 12.1.1.1:10;
217         vrf-target target:64512:10;
218     }
219 }
220 protocols {
221     bgp {
222         group IBGP {
223             type internal;
224             local-address 12.1.1.1;
225             family evpn {
226                 signaling;
227             }
228             neighbor 12.1.1.2;
229         }
230     }
231     isis {
232         interface lo0.0 {
233             passive;
234         }
235         interface ae0.0 {
236             level 1 disable;
237             point-to-point;
238         }
239         level 1 disable;
240         level 2 wide-metrics-only;
241     }
242     lldp {
243         interface et-0/1/4;
244         interface ae0;
245     }
246 }
247 routing-options {
248     router-id 12.1.1.1;
249     autonomous-system 64512;
250     static {
251         route 12.1.1.2/32 next-hop st0.1;
252     }
253     forwarding-table {
254         export pplb;
255     }
256 }

```

1.9.2 PE2 Configuration

```

1 regress@PE2> show configuration groups IPSec_EVPN_VxLAN |no-more
2 chassis {
3     aggregated-devices {
4         ethernet {
5             device-count 30;
6         }
7     }
8     fpc 0 {
9         pic 1 {
10             inline-services;
11         }
12     }
13     network-services enhanced-ip;
14 }
15 services {
16     service-set SS-EVPN-VxLAN {
17         next-hop-service {
18             inside-service-interface si-0/1/0.0;
19             outside-service-interface si-0/1/0.1;
20         }
21         ipsec-vpn ipsec_evpn_vxlan;
22     }
23 }
24 security {
25     ike {
26         proposal ike_proposal {
27             description "IKE Proposal at PE2";
28             authentication-method pre-shared-keys;
29         }

```

```

30     policy ike_policy {
31         mode main;
32         proposals ike_proposal;
33         pre-shared-key ascii-text "$9$fQ3/u0RlK8CtK8X7sYfTz601leMWXNs2"; ## SECRET-DATA
34     }
35     gateway ike_gw {
36         ike-policy ike_policy;
37         address 13.10.1.1;
38         external-interface ae1;
39         local-address 13.10.2.2;
40         version v2-only;
41     }
42 }
43 ipsec {
44     proposal ipsec_proposal {
45         extended-sequence-number;
46         description "IPSec Proposal at PE2";
47         protocol esp;
48         encryption-algorithm aes-256-gcm;
49     }
50     policy ipsec_policy {
51         proposals ipsec_proposal;
52     }
53     vpn ipsec_evpn_vxlan {
54         bind-interface st0.1;
55         copy-outer-dscp;
56         ike {
57             gateway ike_gw;
58             anti-replay-window-size 4096;
59             ipsec-policy ipsec_policy;
60         }
61         establish-tunnels immediately;
62     }
63 }
64 }
65 interfaces {
66     lo0 {
67         unit 0 {
68             family inet {
69                 address 12.1.1.2/32;
70             }
71             family iso {
72                 address 49.0002.0120.0100.1002.00;
73             }
74             family inet6 {
75                 address 2002:12:1:1::2/128;
76             }
77         }
78     }
79     si-0/1/0 {
80         mtu 9192;
81         unit 0 {
82             family inet;
83             family inet6;
84             service-domain inside;
85         }
86         unit 1 {
87             family inet;
88             family inet6;
89             service-domain outside;
90         }
91     }
92     st0 {
93         unit 1 {
94             family inet {
95                 mtu 9192;
96                 address 15.1.1.2/32;
97             }
98             family inet6 {
99                 mtu 9192;
100            }
101        }
102    }
103     et-0/0/4 {
104         description "towards CE2 at IXIA";
105         flexible-vlan-tagging;
106         mtu 9192;
107         encapsulation flexible-ethernet-services;
108         unit 10 {
109             encapsulation vlan-bridge;
110             vlan-id 10;

```

```

111     }
112     unit 20 {
113         encapsulation vlan-bridge;
114         vlan-id 20;
115     }
116 }
117 ael {
118     description "bundle to P1";
119     mtu 9192;
120     aggregated-ether-options {
121         lacp {
122             active;
123             periodic fast;
124         }
125     }
126     unit 0 {
127         family inet {
128             address 13.10.2.2/24;
129         }
130         family iso;
131         family mpls;
132     }
133 }
134 et-0/0/10 {
135     gige-ether-options {
136         802.3ad ael;
137     }
138 }
139 et-0/0/11 {
140     gige-ether-options {
141         802.3ad ael;
142     }
143 }
144 irb {
145     mtu 9192;
146     unit 10 {
147         virtual-gateway-accept-data;
148         family inet {
149             address 13.11.1.2/24 {
150                 virtual-gateway-address 13.11.1.254;
151             }
152         }
153     }
154     unit 20 {
155         virtual-gateway-accept-data;
156         family inet {
157             address 13.11.2.2/24 {
158                 virtual-gateway-address 13.11.2.254;
159             }
160         }
161     }
162 }
163 }
164 policy-options {
165     policy-statement pplb {
166         term pplb {
167             then {
168                 load-balance per-flow;
169             }
170         }
171     }
172     policy-statement nhs {
173         term nhs {
174             then {
175                 next-hop self;
176             }
177         }
178     }
179 }
180 routing-instances {
181     evpn_vxlan_1 {
182         instance-type mac-vrf;
183         protocols {
184             evpn {
185                 encapsulation vxlan;
186                 default-gateway no-gateway-community;
187             }
188         }
189         vtep-source-interface lo0.0;
190         bridge-domains {
191             V_10 {

```

```

192         vlan-id 10;
193         interface et-0/0/4.10;
194         routing-interface irb.10;
195         vxlan {
196             vni 10010;
197         }
198     }
199     V_20 {
200         vlan-id 20;
201         interface et-0/0/4.20;
202         routing-interface irb.20;
203         vxlan {
204             vni 10020;
205         }
206     }
207 }
208 service-type vlan-aware;
209 route-distinguisher 12.1.1.2:1;
210 vrf-target target:64512:1;
211 }
212 vrf_1 {
213     instance-type vrf;
214     interface irb.10;
215     interface irb.20;
216     route-distinguisher 12.1.1.2:10;
217     vrf-target target:64512:10;
218 }
219 }
220 protocols {
221     bgp {
222         group IBGP {
223             type internal;
224             local-address 12.1.1.2;
225             family evpn {
226                 signaling;
227             }
228             neighbor 12.1.1.1;
229         }
230     }
231     isis {
232         interface lo0.0 {
233             passive;
234         }
235         interface ae1.0 {
236             level 1 disable;
237             point-to-point;
238         }
239         level 1 disable;
240         level 2 wide-metrics-only;
241     }
242     lldp {
243         interface et-0/1/4;
244         interface ae1;
245     }
246 }
247 routing-options {
248     router-id 12.1.1.2;
249     autonomous-system 64512;
250     static {
251         route 12.1.1.1/32 next-hop st0.1;
252     }
253     forwarding-table {
254         export pplb;
255     }
256 }

```

1.9.3 P1 Configuration

```

1 regress@P1> show configuration groups IPSec_EVPN_VxLAN |no-more
2 chassis {
3     aggregated-devices {
4         ethernet {
5             device-count 30;
6         }
7     }
8     network-services enhanced-ip;
9 }

```

```

10 interfaces {
11   ae0 {
12     description "Bundle to PE1";
13     mtu 9192;
14     aggregated-ether-options {
15       lacp {
16         active;
17         periodic fast;
18       }
19     }
20     unit 0 {
21       family inet {
22         address 13.10.1.2/24;
23       }
24       family iso;
25     }
26   }
27   et-0/2/0 {
28     gigether-options {
29       802.3ad ae0;
30     }
31   }
32   et-0/2/1 {
33     gigether-options {
34       802.3ad ae0;
35     }
36   }
37   lo0 {
38     unit 0 {
39       family inet {
40         address 12.1.1.3/32;
41       }
42       family iso {
43         address 49.0002.0120.0100.1003.00;
44       }
45       family inet6 {
46         address 2002:12:1:1::3/128;
47       }
48     }
49   }
50   ae1 {
51     description "Bundle to PE2";
52     mtu 9192;
53     aggregated-ether-options {
54       lacp {
55         active;
56         periodic fast;
57       }
58     }
59     unit 0 {
60       family inet {
61         address 13.10.2.1/24;
62       }
63       family iso;
64     }
65   }
66   et-0/0/0 {
67     gigether-options {
68       802.3ad ae1;
69     }
70   }
71   et-0/2/2 {
72     gigether-options {
73       802.3ad ae1;
74     }
75   }
76 }
77 policy-options {
78   policy-statement pplb {
79     term pplb {
80       then {
81         load-balance per-flow;
82       }
83     }
84   }
85 }
86 routing-options {
87   router-id 12.1.1.3;
88   autonomous-system 64512;
89   forwarding-table {
90     export pplb;

```

```

91     }
92 }
93 protocols {
94     isis {
95         interface lo0.0 {
96             passive;
97         }
98         interface ae0.0 {
99             level 1 disable;
100             point-to-point;
101         }
102         interface ae1.0 {
103             level 1 disable;
104             point-to-point;
105         }
106         level 1 disable;
107         level 2 wide-metrics-only;
108     }
109     lldp {
110         interface ae0;
111         interface ae1;
112     }
113 }

```

1.9.4 EVPN-VXLAN Verification

```

1 regress@PE1> show bgp summary
2 Threading mode: BGP I/O
3 Default eBGP mode: advertise - accept, receive - accept
4 Groups: 1 Peers: 1 Down peers: 0
5 Table Tot Paths Act Paths Suppressed History Damp State Pending
6 bgp.evpn.0
7      13 13 0 0 0 0
8 Peer AS InPkt OutPkt OutQ Flaps Last Up/Dwn State|#Active/Received/Accepted/Damped...
9 12.1.1.2 64512 3041 3047 0 0 22:39:39 Establ
10 bgp.evpn.0: 13/13/13/0
11 evpn_vxlan_1.evpn.0: 13/13/13/0
12 __default_evpn__.evpn.0: 0/0/0/0

```

The MAC-VRF database capturing the mac and IP bindings are as follows:

```

1 regress@PE1> show mac-vrf routing database
2 Instance: evpn_vxlan_1
3 VLAN DomainId MAC address Active source Timestamp IP address
4 10010 00:00:5e:00:01:01 05:00:00:fc:00:00:27:1a:00 Nov 27 20:54:39 13.11.1.254
5 10010 00:11:01:00:00:01 et-0/2/10.10 Nov 27 20:52:40
6 10010 00:12:01:00:00:01 12.1.1.2 Nov 27 20:54:39
7 10010 a4:7f:1b:ce:2a:91 12.1.1.2 Nov 27 20:54:39 13.11.1.2
8 10010 d4:99:6c:92:48:fc irb.10 Nov 27 20:52:40 13.11.1.1
9 10020 00:00:5e:00:01:01 05:00:00:fc:00:00:27:24:00 Nov 27 20:54:39 13.11.2.254
10 10020 a4:7f:1b:ce:2a:91 12.1.1.2 Nov 27 20:54:39 13.11.2.2
11 10020 d4:99:6c:92:48:fc irb.20 Nov 27 20:52:40 13.11.2.1
12
13 regress@PE1> show mac-vrf forwarding mac-ip-table
14
15 MAC IP flags (S - Static, D - Dynamic, L - Local , R - Remote, Lp - Local Proxy,
16 Rp - Remote Proxy, K - Kernel, RT - Dest Route, (N)AD - (Not) Advt to remote,
17 RE - Re-ARP/ND, RO - Router, OV - Override, Ur - Unresolved, B - Blocked,
18 RTS - Dest Route Skipped, RGW - Remote Gateway, RTF - Dest Route Forced,
19 SC - Static Config, P - Probe, NLC - No Local Config, LD - Local Down)
20 Routing instance : evpn_vxlan_1
21 Bridging domain : V_10
22 IP MAC Flags GBP Logical Active
23 address address Tag Interface source
24 13.11.1.254 00:00:5e:00:01:01 S,K irb.10
25 13.11.1.2 a4:7f:1b:ce:2a:91 SR,K,RT vtep.32769 12.1.1.2
26 13.11.1.1 d4:99:6c:92:48:fc S,K irb.10
27
28 MAC IP flags (S - Static, D - Dynamic, L - Local , R - Remote, Lp - Local Proxy,
29 Rp - Remote Proxy, K - Kernel, RT - Dest Route, (N)AD - (Not) Advt to remote,
30 RE - Re-ARP/ND, RO - Router, OV - Override, Ur - Unresolved, B - Blocked,
31 RTS - Dest Route Skipped, RGW - Remote Gateway, RTF - Dest Route Forced,
32 SC - Static Config, P - Probe, NLC - No Local Config, LD - Local Down)
33 Routing instance : evpn_vxlan_1
34 Bridging domain : V_20
35 IP MAC Flags GBP Logical Active

```

```

36 address address Tag Interface source
37 13.11.2.254 00:00:5e:00:01:01 S,K irb.20
38 13.11.2.2 a4:7f:1b:ce:2a:91 SR,K,RT vtep.32769 12.1.1.2
39 13.11.2.1 d4:99:6c:92:48:fc S,K irb.20

```

The IPSec stats captures the packets encrypted and decrypted

```

1 regress@PE1> show security ipsec statistics
2 ESP Statistics:
3   Encrypted bytes: 21879502721116
4   Decrypted bytes: 24148488261124
5   Encrypted packets: 40517598197
6   Decrypted packets: 40517598266
7 AH Statistics:
8   Input bytes: 0
9   Output bytes: 0
10  Input packets: 0
11  Output packets: 0
12 Errors:
13   AH authentication failures: 0, Replay errors: 0
14   ESP authentication failures: 0, ESP decryption failures: 0
15   Bad headers: 0, Bad trailers: 0

```

1.10 Scale and Performance of Inline IPSec

1.10.1 Inline IPSec Scale

Trio 6 based MX platforms support 2,000 Inline IPSec Tunnels per chassis. Each PFE supports 1,000 tunnels. The throughput supported is 600Gbps per Trio 6 ASIC. The scale of the tunnels is covered with inline IPSec's traffic selection mode. In the traffic selector mode, we explicitly map the source and destination addresses in the IPSec VPN and there is no requirement to map the route to st0 via static route. IKED adds a route based on the remote-ip, instead of user adding a static route or BGP/IGP adding routes. Traffic Selector mode provides more granularity to steer the traffic of specific interest to a IPVPN tunnel.

```

1 security {
2   ipsec {
3     vpn ipsec_vpn_v4_srv6 {
4       bind-interface st0.1;
5       copy-outer-dscp;
6       ike {
7         gateway ike_gw;
8         anti-replay-window-size 4096;
9         ipsec-policy ipsec_policy;
10      }
11      traffic-selector ts1 {
12        local-ip 50.1.0.0/16;
13        remote-ip 60.1.0.0/16;
14      }
15      establish-tunnels immediately;
16    }
17    vpn ipsec_vpn_v6_srv6 {
18      bind-interface st0.2;
19      copy-outer-dscp;
20      ike {
21        gateway ike_gw;
22        anti-replay-window-size 4096;
23        ipsec-policy ipsec_policy;
24      }
25      traffic-selector ts2 {
26        local-ip 3000:50:1::/48;
27        remote-ip 3000:60:1::/48;
28      }
29      establish-tunnels immediately;
30    }
31  }
32 }

```

In the test topology 2,000 virtual routers are created in the CE which forms the ebgp peering with the PE

devices. Inline IPsec tunnel is setup between the respective CE's with traffic selector mode, 1,000 tunnels created from PFE0 (si-2/1/0) and 1,000 tunnels created from PFE1 (si-2/1/2)

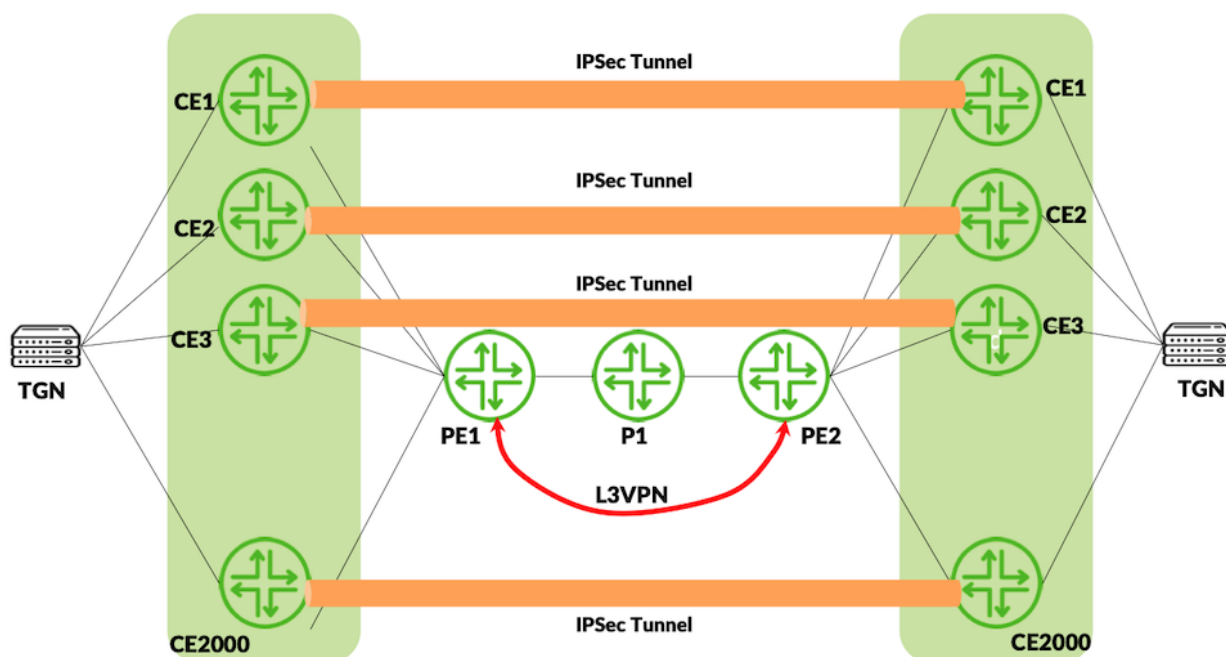


Figure 10: Test topology

```

1 {master}
2 regress@ce1> show security ike security-associations |match UP | count
3 Count: 2000 lines
4
5 regress@ce1> show security ipsec security-associations
6 Total active tunnels: 2000 Total IPsec sas: 2000
7 ID Algorithm SPI Life:sec/kb Mon lsys Port Gateway
8 <508816 ESP:aes-gcm-256/aes256-gcm 0x78cb23e5 799/ unlim - root 500 173.8.209.1
9 >508816 ESP:aes-gcm-256/aes256-gcm 0x8d71f1b3 799/ unlim - root 500 173.8.209.1
10 <506755 ESP:aes-gcm-256/aes256-gcm 0x90774003 1244/ unlim - root 500 173.5.49.1
11 >506755 ESP:aes-gcm-256/aes256-gcm 0xd5045f3c 1244/ unlim - root 500 173.5.49.1
12 ...
13 ...
14 <507441 ESP:aes-gcm-256/aes256-gcm 0x49a8adb8 2019/ unlim - root 500 173.3.176.1
15 >507441 ESP:aes-gcm-256/aes256-gcm 0x1826ec5c 2019/ unlim - root 500 173.3.176.1
16 <507732 ESP:aes-gcm-256/aes256-gcm 0xe397fe4b 1963/ unlim - root 500 173.3.53.1
17 >507732 ESP:aes-gcm-256/aes256-gcm 0x9bfd9333 1963/ unlim - root 500 173.3.53.1
18
19 {master}
20 regress@ce1>
21
22 {master}
23 regress@ce1> show interfaces st0 terse |match inet | count |except inet6
24 Count: 2000 lines
25
26 {master}
27 regress@ce1> show interfaces terse si-2/1/0 |except inet6 | count
28 Count: 2002 lines
29
30 {master}
31 regress@ce1> show interfaces terse si-2/1/2 |except inet6 | count
32 Count: 2002 lines
33
34 {master}
35 regress@ce1> show security ipsec statistics
36 ESP Statistics:
37 Encrypted bytes: 2596055034
38 Decrypted bytes: 2739372792
39 Encrypted packets: 2654453
40 Decrypted packets: 2654431
41 AH Statistics:
42 Input bytes: 0

```

```
43  Output bytes: 0
44  Input packets: 0
45  Output packets: 0
46  Errors:
47  AH authentication failures: 0, Replay errors: 0
48  ESP authentication failures: 0, ESP decryption failures: 0
49  Bad headers: 0, Bad trailers: 0
```

1.11 Inline IPSec Performance

MX304 inline IPSec provides 300Gbps bandwidth per PFE and IPv4 and IPv6 performance on a single PFE is captured as follows. Total throughput per Trio 6 is 600Gbps

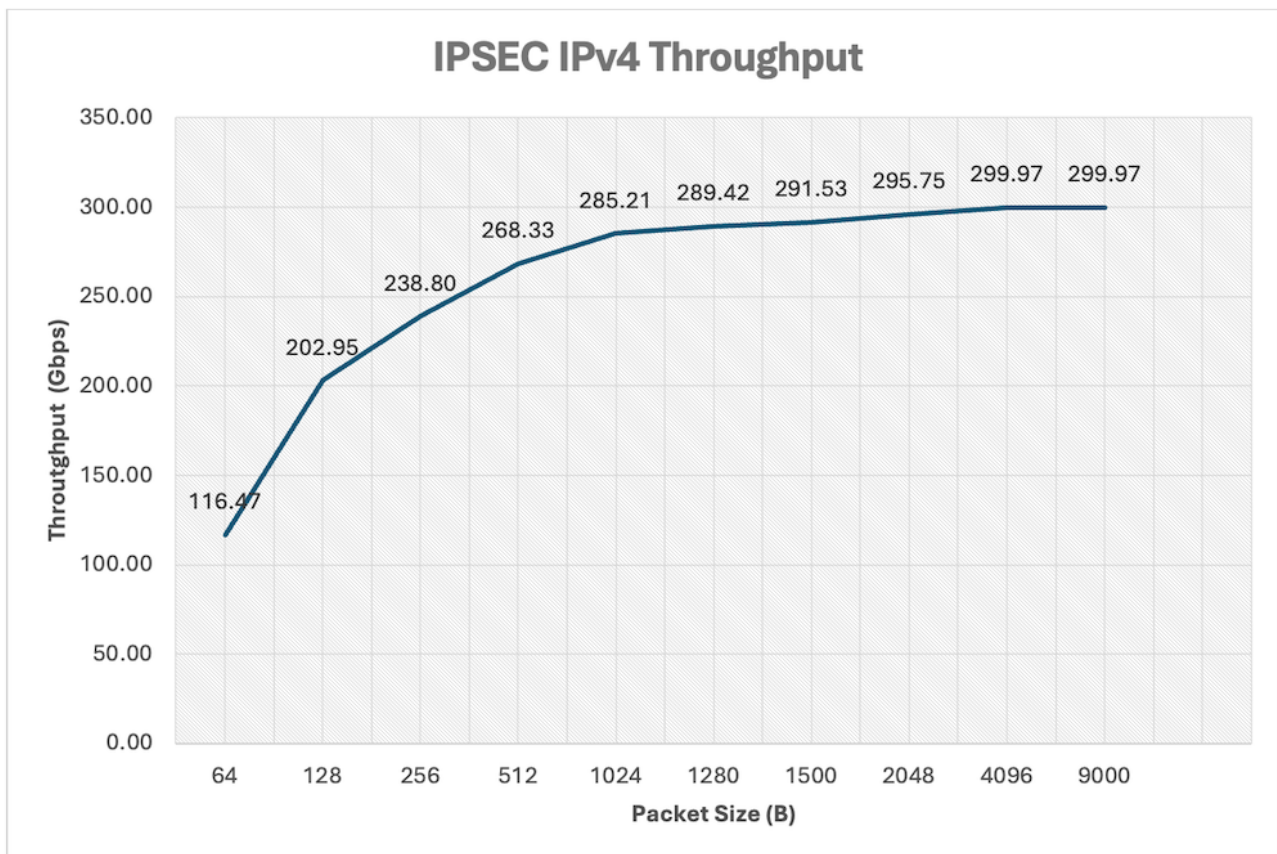


Figure 11: IPsec IPv4 Throughput

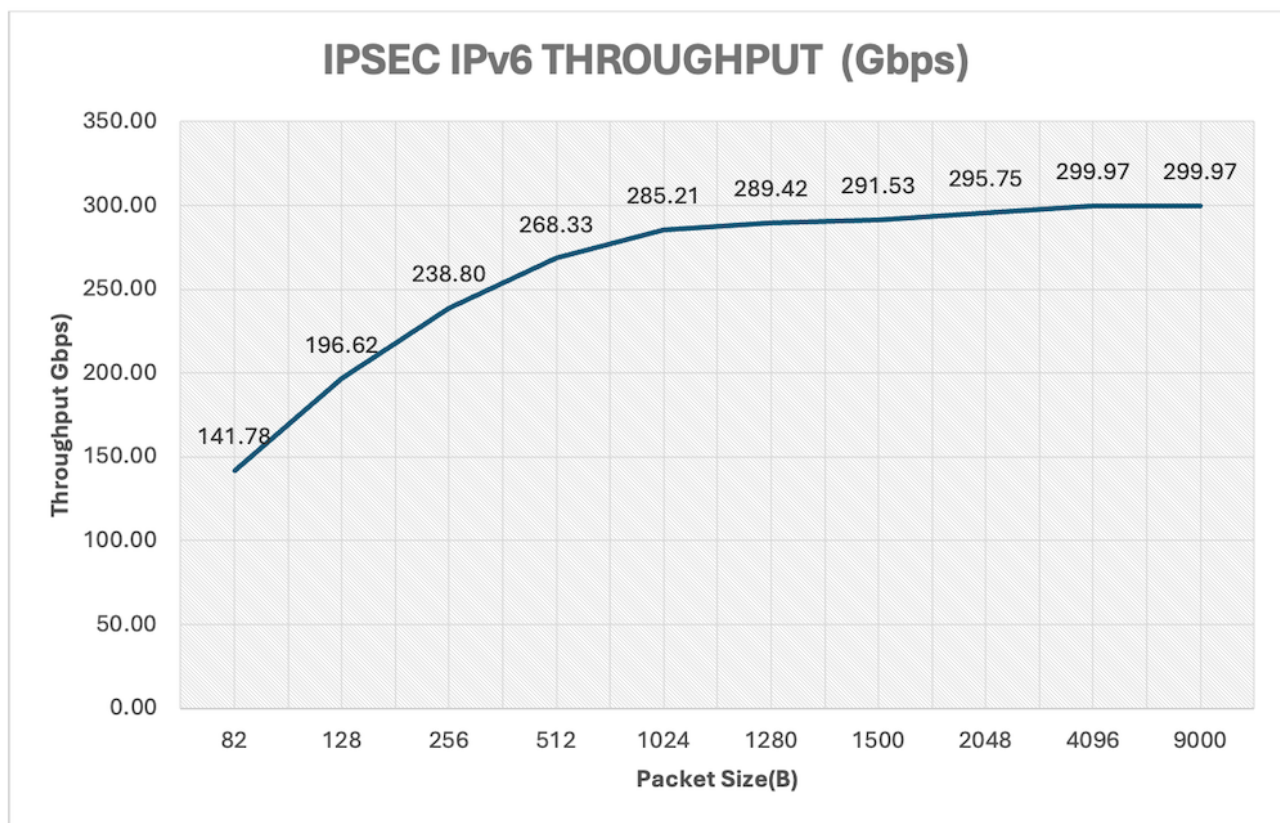


Figure 12: IPsec IPv6 Throughput

1.12 Useful links

- <https://www.juniper.net/documentation/us/en/software/junos/interfaces-adaptive-services/topics/topic-map/inline-ipsec.html>

1.13 Glossary

- AES: Advanced Encryption Standard
- CPU: Central Processing Unit, the general-purpose processor of the router (implied, contrasted with PFE)
- DPD: Dead Peer Detection, a mechanism to verify the liveness of the IKE peer to prevent black-holing of IPsec traffic.
- ESP: Encapsulating Security Payload, the IPsec protocol used for encrypting and ensuring the integrity of tunneled data.
- IKE: Internet Key Exchange, the protocol used to negotiate cryptographic keys and establish Security Associations (SAs) for IPsec.
- IKEv2: Version 2 of IKE, the modern version of the key-management protocol supported by inline IPsec on MX platforms.
- SA: Security Association, a one-way agreement between two endpoints that defines the IPsec protections (encryption, integrity, keys, lifetime) for traffic.
- SI: Service Interface, a virtual interface on the PFE used as the anchor for inline services (like IPsec) without requiring external service cards.

- SPI: Security Parameter Index, a unique identifier for an SA on a host or router.
- USF: Unified Services Framework, a configuration mode on MX routers that must be enabled to support inline IPsec (instead of relying on external service cards).

1.14 Acknowledgments

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